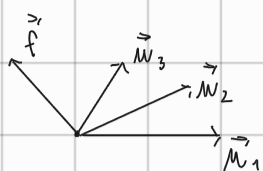


RAPPRESENTAZIONE MATRICIALE DI OPERATORI e FUNZIONI D'ONDA:

Spaz. Vett. $\mathbb{C}^3$



$$\vec{m}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{m}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{m}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

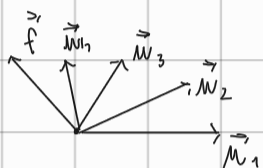
$$\vec{f} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = |\vec{f}\rangle \Leftrightarrow \langle \vec{f} | = |\vec{f}\rangle^{*T}$$

$$\langle \vec{g} | \vec{f} \rangle = \langle \vec{f} | \vec{g} \rangle^*$$

$$|\vec{m}_n\rangle \Rightarrow |\vec{f}\rangle = \sum_n c_n |\vec{m}_n\rangle, \quad \langle \vec{m}_n | \vec{m}_m \rangle = \delta_{nm}$$

MATRICI $\hat{M}$

Spazi Funz.: $\mathbb{L}^2$



$$|f\rangle \Leftrightarrow \langle f |^*$$

$$\langle f | g \rangle = \int f^* g dx$$

$$\{m_n\} \Leftrightarrow |f\rangle = \sum_n c_n |m_n\rangle$$

Stessa struttura

OPERATORI $\hat{O}$

(unisco i due)

=> SPAZIO ASTRATTO dei KET H

↳ Spazio vett. complessa $|f\rangle$

$$|\chi\rangle \in |f\rangle \Rightarrow |\chi\rangle = \alpha |\psi\rangle + \beta |f\rangle \quad \text{LINEARITÀ} \quad \alpha, \beta \in \mathbb{C}$$

$$\text{Ad ogni } |\cdot\rangle \Leftrightarrow \langle \cdot | = |\cdot\rangle^{*T} = |\cdot\rangle^\dagger$$

• Prodotto interno $\Leftrightarrow$ Prodotto scalare

$$\langle f | \psi \rangle = \langle \psi | f \rangle^*$$

• Lo spaz. dei KET si dice completo rispetto all'a norma indotta dal prodotto scalare.

$$\|\psi\| = \sqrt{\langle \psi | \psi \rangle}$$

$$\cdot \left\{ |n\rangle \right\}^{\text{BASE}} \text{ t.c. } |\psi\rangle = \sum_n c_n |n\rangle$$

(finalmente non scelgo una base precisa
: ket rimangono ogg. astratti)

SCELGO UNA BASE: $|m_i\rangle$

$$|f\rangle = \sum_i c_i |m_i\rangle \quad c_i = \langle m_i | f \rangle$$

$$|\vec{v}\rangle = 2|\vec{m}_x\rangle + 3|\vec{m}_y\rangle + 4|\vec{m}_z\rangle \Rightarrow |\vec{v}\rangle = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

$$|\psi\rangle = \sqrt{\frac{3}{5}}|Y_1^1\rangle + \sqrt{\frac{1}{5}}|Y_1^0\rangle + \sqrt{\frac{1}{5}}|Y_1^{-1}\rangle \quad \begin{cases} l=1 \\ m_l=1 \end{cases}$$

$$|\psi\rangle = \begin{pmatrix} \sqrt{3/5} \\ \sqrt{1/5} \\ \sqrt{1/5} \end{pmatrix} \quad \begin{array}{l} \text{1a vett. dei coef. SOSTITUIRE} \\ \text{una BASE dettata. univ. vec.} \\ \text{una BASE} \end{array}$$

$$\Rightarrow |\psi\rangle \rightarrow \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{pmatrix} \quad \text{n comp. della base}$$

$$|\varphi\rangle = |\hat{O}\psi\rangle = \{ |w_n\rangle \} = \hat{O} \sum_j c_j |w_j\rangle = \sum_j c_j \hat{O} |w_j\rangle$$

$$\Rightarrow \langle w_i | \varphi \rangle = \langle w_i | \hat{O} \psi \rangle = \langle w_i | \sum_j c_j \hat{O} |w_j\rangle \Rightarrow b_i = \langle w_i | \varphi \rangle = \sum_j c_j \langle w_i | \hat{O} |w_j\rangle$$

Proiezione di $|\varphi\rangle$ sul ket di base w_i

$$b_i = (\phi)_i = \sum_j c_j \langle w_i | \hat{O} |w_j\rangle$$

$$O_{ij} = \langle w_i | \hat{O} |w_j\rangle$$

$$\Rightarrow b_i = (\phi)_i = \sum_j O_{ij} c_j$$

$$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \end{pmatrix} = \begin{pmatrix} (\hat{O}\psi)_1 \\ \vdots \\ (\hat{O}\psi)_i \\ \vdots \end{pmatrix} = \begin{pmatrix} O_{11} & O_{12} & \dots & O_{1j} & \dots \\ \vdots & \vdots & & \vdots & \dots \\ O_{i1} & & & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_j \\ \vdots \end{pmatrix}$$

↓
Vett. proiezione $|\varphi\rangle$
su $\langle w$

$$b_1 = \sum_j O_{1j} c_j = O_{11}c_1 + O_{12}c_2 + \dots$$

$|\varphi\rangle = |\hat{O}\psi\rangle \Leftrightarrow \varphi = \hat{O}\psi \Rightarrow$ Ho trasf. questo operaz. diff. in una matrice per un vettore

Rappr. di $|\psi\rangle$
sulla base $|w_i\rangle$

MATRICE di $\hat{O}$ sulla
base $|w_i\rangle$

Rappresent. di $|\psi\rangle$ sulla
base $|w_i\rangle$

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \end{pmatrix} = \begin{pmatrix} (\hat{O}\psi)_1 \\ \vdots \\ (\hat{O}\psi)_i \\ \vdots \end{pmatrix} = \begin{pmatrix} O_{11} & O_{12} & \dots & O_{1j} & \dots \\ \vdots & \vdots & & \vdots & \dots \\ \vdots & \vdots & & \vdots & \dots \\ O_{i1} & & & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \end{pmatrix}$$

BASE N $|w_i\rangle \Rightarrow \hat{O} \ N \times N$

$$|\psi\rangle = \sum_i c_i |w_i\rangle = \sum_i |w_i\rangle \langle w_i | \psi \rangle = \sum_i \underbrace{(|w_i\rangle \langle w_i|)}_{\mathbb{I} \text{ (op. identit.)}} |\psi\rangle$$

$\langle w_i | \hat{O} | w_j \rangle \Rightarrow$ Elemento di matr. di $\hat{O}$

$\mathbb{I}$ (op. identit.)

$$\langle w_i | \hat{O} \hat{P} | w_j \rangle = \sum_k \langle w_i | \hat{O} | w_k \rangle \langle w_k | \hat{P} | w_j \rangle = \sum_k O_{ik} P_{kj}$$

$$(\hat{O} \hat{P})_{ij} = \sum_k O_{ik} P_{kj}$$

$$\begin{pmatrix} (OP)_{11} & (OP)_{12} & \dots \\ \vdots & \vdots & \\ (OP)_{i1} & (OP)_{ij} & \dots \end{pmatrix} = \begin{pmatrix} O_{11} & \dots & O_{1j} & \dots \\ \vdots & & \vdots & \\ O_{i1} & \dots & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} P_{11} & \dots & P_{1j} & \dots \\ \vdots & & \vdots & \\ P_{i1} & \dots & P_{ij} & \dots \end{pmatrix}$$

$$(OP)_{11} = \sum_k O_{1k} P_{k1} = O_{11} P_{11} + O_{12} P_{21} + O_{13} P_{31} + \dots$$

Il prod. tra due op.
è diventato un prod. tra
matrici

$$(\hat{O}^\dagger)_{ji} = \langle w_i | \hat{O}^\dagger | w_j \rangle = \langle \hat{O} w_i | w_j \rangle = \langle w_j | \hat{O} | w_i \rangle^* = O_{ji}^*$$

1) $\hat{O} \psi \Rightarrow [\hat{O}] [\psi]$ ^{matrice}

2) $\hat{O} \hat{P} \Rightarrow [\hat{O}] [\hat{P}]$

3) $\hat{O}^\dagger \Rightarrow O_{ji}^*$

① $|m_j\rangle = \text{SET di AUTOSTATI di } \hat{O} \Rightarrow \langle m_i | \hat{O} | m_j \rangle = 0_j \delta_{ij}$

② $\hat{O} \Rightarrow \text{SPETTRO di AUTOSTATI discreto} \Rightarrow \hat{O} \Rightarrow \text{MAT } N \times N$

$\hat{O} \Rightarrow \text{SPETTRO CONTINUO} \Rightarrow \text{MAT. } \times \text{ di } \hat{O} \text{ (inf. non numerabile)}$

$\hat{H} \Rightarrow \text{SCELGO una base qualunque} \Rightarrow \langle m_i | \hat{H} | m_j \rangle$

$$\begin{pmatrix} H_{11} & H_{12} & \dots \\ \vdots & \vdots & \vdots \end{pmatrix}$$

sono mat. che mi interessano $\Rightarrow$ blocchi

(mi interessano solo alcuni valori)

Es. di rappresent. matriciale:

$\Rightarrow$ Matrici di $\hat{L}_z$ per $l=1$ $Y_1^1 \Rightarrow |1,1\rangle$, $Y_1^0 \Rightarrow |1,0\rangle$, $Y_1^{-1} \Rightarrow |1,-1\rangle$

$|\psi\rangle = \sum_{l,m} c_{l,m} |l,m\rangle \Rightarrow |\psi\rangle = \sum_m c_m |\bar{l}, m\rangle$

$|\psi\rangle = \alpha |1,1\rangle + \beta |1,0\rangle + \gamma |1,-1\rangle \Rightarrow \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$

$\hat{L}_z |l,m\rangle = \hbar m |l,m\rangle \Rightarrow \hat{L}_z |1,m\rangle = \hbar m |1,m\rangle$

$\langle 1, m' | \hat{L}_z |1,m\rangle = \hbar m \delta_{m,m'} \Leftrightarrow \langle 1, m' | \hbar m |1,m\rangle$

$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \Rightarrow L_z = \begin{pmatrix} \langle 1,1 | \hat{L}_z |1,1\rangle & \langle 1,1 | \hat{L}_z |1,0\rangle & \langle 1,1 | \hat{L}_z |1,-1\rangle \\ \langle 1,0 | \hat{L}_z |1,1\rangle & \langle 1,0 | \hat{L}_z |1,0\rangle & \langle 1,0 | \hat{L}_z |1,-1\rangle \\ \langle 1,-1 | \hat{L}_z |1,1\rangle & \langle 1,-1 | \hat{L}_z |1,0\rangle & \langle 1,-1 | \hat{L}_z |1,-1\rangle \end{pmatrix} =$

$= \begin{pmatrix} \hbar & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\hbar \end{pmatrix} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$

Trovo $\hat{L}_x$ e $\hat{L}_y$

$\hat{L}_+ |l,m\rangle = \hbar \sqrt{l(l+1) - m(m+1)} |l, m+1\rangle$

$\hat{L}_+^\dagger = \hat{L}_-$

$\hat{L}_- |l,m\rangle = \hbar \sqrt{l(l+1) - m(m-1)} |l, m-1\rangle$

$$\bullet \langle l, m' | \hat{L}_+ | l, m \rangle = \sqrt{l(l+1) - m(m+1)} \hbar \delta_{m', m+1}$$

$$\bullet \langle l, m' | \hat{L}_- | l, m \rangle = \sqrt{l(l+1) - m(m-1)} \hbar \delta_{m', m-1}$$

$$\hat{L}_+ = \begin{pmatrix} \langle 1, 1 | \hat{L}_+ | 1, 1 \rangle & \langle 1, 1 | \hat{L}_+ | 1, 0 \rangle & \langle 1, 1 | \hat{L}_+ | 1, -1 \rangle \\ \langle 1, 0 | \hat{L}_+ | 1, 1 \rangle & \langle 1, 0 | \hat{L}_+ | 1, 0 \rangle & \langle 1, 0 | \hat{L}_+ | 1, -1 \rangle \\ \langle 1, -1 | \hat{L}_+ | 1, 1 \rangle & \langle 1, -1 | \hat{L}_+ | 1, 0 \rangle & \langle 1, -1 | \hat{L}_+ | 1, -1 \rangle \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \hbar$$

$$\hat{L}_- = \hbar \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

$$\hat{L}_x = \frac{1}{2} (\hat{L}_+ + \hat{L}_-) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{L}_y = \frac{1}{2i} (\hat{L}_+ - \hat{L}_-) = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\Gamma l=2 \Rightarrow m_l = -2, -1, 0, 1, 2$$

$$S = \frac{1}{2} \quad m_s = \pm \frac{1}{2} \quad |\uparrow\rangle, |\downarrow\rangle \quad |\chi\rangle = c_1 |\uparrow\rangle + c_2 |\downarrow\rangle \Rightarrow |\chi\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

OPERAT. A SPECTRO CONT.:

$$|\psi\rangle = \sum_i c_i |m_i\rangle \quad |m_i\rangle \Rightarrow |x\rangle$$

$$\downarrow$$

$$|\psi\rangle = \int c(x) |x\rangle dx \quad , \quad \langle x | \psi \rangle = \int dx' c(x') \underbrace{\langle x | x' \rangle}_{\delta(x-x')}$$

$$\langle x | \psi \rangle = \int dx' c(x') \delta(x-x') = c(x)$$

$|c(x)|^2 \Rightarrow$ Dens. probab. corrispond. al ket $|x\rangle$

$$\Rightarrow |c(x)|^2 = |\psi(x)|^2 \Rightarrow c(x) = \psi(x) \Rightarrow \boxed{\psi(x) = \langle x | \psi \rangle}$$

$$|\psi\rangle \Rightarrow \begin{pmatrix} \psi(x_1) \\ \psi(x_2) \\ \vdots \\ \psi(x_n) \end{pmatrix} \quad \text{infin. non numerabile}$$

TEORIA DELLE PERTURB. NON DIPENDENTI DAL TEMPO: STATI NON DEGENERI

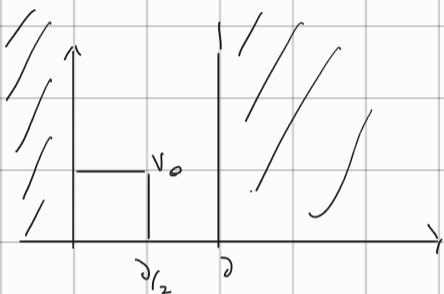
$\hat{H}^{(0)}$ HAMILTON. IMPERTURBATA } LO RISORNO ESATT. $\Rightarrow \hat{H}^{(0)} w_n^{(0)} = E_n^{(0)} w_n^{(0)}$

VALORE ESATT. $w_n^{(0)} E_n^{(0)}$

$$\langle w_n^{(0)} | w_m^{(0)} \rangle = \int w_n^{(0)} w_m^{(0)} d\tau = \int_{-l/2}^{l/2} \delta_{n,m} \quad , \quad \langle w_n^{(0)} | \hat{H}^{(0)} | w_m^{(0)} \rangle = E_n^{(0)} \int_{-l/2}^{l/2} \delta_{n,m}$$

E_b : BUCIA POT. INF.

$$w_n^{(0)}(x) = \sqrt{\frac{2}{l}} \sin\left(\frac{n\pi x}{l}\right) \quad n = 1, 2, 3 \quad E_n^{(0)} = \frac{\pi^2 \hbar^2 n^2}{2m_e l^2}$$



AGGIUNGO V_0

$$\hat{V}(x) = \begin{cases} V_0 & 0 \leq x \leq l/2 \\ 0 & \text{altrove} \end{cases}$$

PERTURBAZIONE $\hat{H}' = \hat{V}(x)$

SISTEMA PERTURB.: $\hat{H} = \hat{H}^{(0)} + \lambda \hat{H}'$

$$\hookrightarrow \hat{H} w_n = E_n w_n \Rightarrow \text{NON SI POSSO CASCARE}$$

$\lambda \rightarrow$ Parametro piccolo $\lambda \ll 1 \Rightarrow \lambda \hat{H}' \ll \hat{H}^{(0)} \Rightarrow \hat{V}_0$ piccolo rispetto al sistema

$$\Rightarrow \begin{cases} w_n = w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots \\ E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \end{cases}$$

$\lambda^k E_n^{(k)} \rightarrow$ ORDINE della CORREZIONE
Pot. di λ

$$\hat{H} w_n = E_n w_n \Rightarrow$$

$$[\hat{H}^{(0)} + \lambda \hat{H}'] [w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots] = [E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots] [w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots]$$

$\rightarrow$ ESPRIMO SOLO TERMINI ALLA II^a POT. in λ

$$\hat{H}^{(0)} w_n^{(0)} + \lambda \hat{H}^{(0)} w_n^{(1)} + \lambda^2 \hat{H}^{(0)} w_n^{(2)} + \lambda \hat{H}' w_n^{(0)} + \lambda^2 \hat{H}' w_n^{(1)} + \dots =$$

$$= E_n^{(0)} w_n^{(0)} + \lambda E_n^{(0)} w_n^{(1)} + \lambda^2 E_n^{(0)} w_n^{(2)} + \lambda E_n^{(1)} w_n^{(0)} + \lambda^2 E_n^{(1)} w_n^{(1)} + \lambda^2 E_n^{(2)} w_n^{(0)} + \dots$$

$\Rightarrow$ RACCOLGO i term. sulle base delle pot. di λ

$$\begin{aligned}
 &= \underbrace{\hat{H}^{(0)} w_n^{(0)}}_{\lambda^0} + \underbrace{\lambda [\hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)}]}_{1^{\text{ste pot}} \lambda} + \underbrace{\lambda^2 [\hat{H}^{(0)} w_n^{(2)} + \hat{H}' w_n^{(1)}]}_{2^{\text{te pot}} \lambda} + \dots = \\
 &= \underbrace{E_n^{(0)} w_n^{(0)}}_{\lambda^0} + \underbrace{\lambda [E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)}]}_{1^{\text{ste pot}} \lambda} + \underbrace{\lambda^2 [E_n^{(0)} w_n^{(2)} + E_n^{(1)} w_n^{(1)} + E_n^{(2)} w_n^{(0)}]}_{2^{\text{te pot}} \lambda} + \dots
 \end{aligned}$$

$$\boxed{\lambda^0} \quad \hat{H}^{(0)} w_n^{(0)} = E_n^{(0)} w_n^{(0)}$$

$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)}$$

$$\boxed{\lambda^2} \quad \hat{H}^{(0)} w_n^{(2)} + \hat{H}' w_n^{(1)} = E_n^{(0)} w_n^{(2)} + E_n^{(1)} w_n^{(1)} + E_n^{(2)} w_n^{(0)}$$

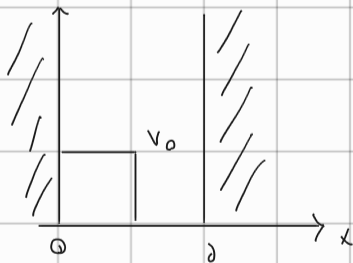
$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)} \quad \text{Multipl. per } w_n^{(0)*} \text{ e integ.}$$

$$\begin{aligned}
 \Rightarrow \langle w_n^{(0)} | \hat{H}^{(0)} | w_n^{(1)} \rangle + \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle &= E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \underbrace{\langle w_n^{(0)} | w_n^{(0)} \rangle}_1 \\
 \hookrightarrow \langle \hat{H}^{(0)} w_n^{(0)} | w_n^{(1)} \rangle &= E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle
 \end{aligned}$$

$$\Rightarrow \cancel{E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle} + \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle = \cancel{E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle} + E_n^{(1)}$$

$$\Rightarrow \boxed{E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle} = \int_{\mathbb{R}} w_n^{(0)*} \hat{H}' w_n^{(0)} dx$$

Es.:



$$\hat{H}^{(0)} = \frac{\hat{p}^2}{2m}$$

$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad n = 1, 2, 3, \dots$$

$$E_n^{(0)} = \frac{\pi^2 \hbar^2 n^2}{2m_e a^2}$$

$$\hat{H}' = V = \begin{cases} V_0 & 0 \leq x \leq a/2 \\ 0 & \text{ALTRONDE} \end{cases}$$

$$\Rightarrow E_n^{(1)} = \int w_n^{(0)*} \hat{H}' w_n^{(0)} dx = \int_0^{a/2} w_n^{(0)*} V_0 w_n^{(0)} dx$$

$$= V_0 \int_0^{a/2} |w_n^{(0)}|^2 dx = V_0 \int_0^{a/2} \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{V_0}{2}$$

$$\Rightarrow E_1^{(1)} = V_0 \int_0^{a/2} \frac{2}{a} \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{V_0}{2}$$

$$E_n = E_n^{(0)} + \lambda \frac{V_0}{2} \Rightarrow E_n = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2} + \frac{V_0}{2}$$

poiché $\lambda = 1$

AUTOVALORE (IMPERTURBATO) CORRETTO APPROSSIMATO

① DOMINIO di $\hat{H}'$ SI DEVE SOVRAPPORRE ad $\psi_n^{(0)}$

② PARITÀ $\psi_n^{(0)}$ E $\hat{H}'$ HANNO PARITÀ DIFFERENTE $\Rightarrow \int dx = 0$

Autofunzioni:

λ'

$$\hat{H}^{(0)} \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

$$\underbrace{[\hat{H}^{(0)} - E_n^{(0)}]}_{\text{OPERAT.}} \psi_n^{(1)} = - \underbrace{[\hat{H}' - E_n^{(1)}]}_{\text{TERMINE NOTO}} \psi_n^{(0)}$$

$\psi_n^{(0)} \Rightarrow$ BASE COMPLETA $\Rightarrow \psi_n^{(1)} = \sum_m c_m \psi_m^{(0)}$ (comb. lineare di tutte le autof. impert.)
(lo posso fare perché $\psi_n^{(1)}$ vive nello stesso spazio di Hilbert)

$$\psi_n^{(0)} \in \text{Ker} \{ \hat{H}^{(0)} - E_n^{(0)} \} \Rightarrow [\hat{H}^{(0)} - E_n^{(0)}] \psi_n^{(0)} = 0$$

Visto che $\psi_n^{(0)} \in \text{Ker}$ posso sottrarlo a $\psi_n^{(1)}$ e il risult. non cambia

$$\Rightarrow \boxed{\psi_n^{(1)} = \sum_{m \neq n} c_m \psi_m^{(0)}}$$

$$E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle \quad [\hat{H}^{(0)} - E_n^{(0)}] w_n^{(1)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \Rightarrow w_n^{(1)} = \sum_{m \neq n} c_m w_m^{(0)}$$

$$[\hat{H}^{(0)} - E_n^{(0)}] \sum_{m \neq n} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \Rightarrow \sum_{m \neq n} c_m \hat{H}^{(0)} w_m^{(0)} =$$

$$- \sum_{m \neq n} E_m^{(0)} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)}$$

$$\Rightarrow \sum_{m \neq n} E_m^{(0)} w_m^{(0)} - \sum_{m \neq n} E_m^{(0)} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)}$$

$$\Rightarrow \sum_{m \neq n} (E_m^{(0)} - E_n^{(0)}) c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \quad \text{Moltip. tutto per } w_\ell^{(0)*}$$

$$\Rightarrow \sum_{m \neq n} (E_m^{(0)} - E_n^{(0)}) c_m \langle w_\ell^{(0)} | w_m^{(0)} \rangle = - \langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \langle w_\ell^{(0)} | w_n^{(0)} \rangle$$

$$\bullet \boxed{\ell = n} \Rightarrow \langle w_\ell^{(0)} | w_n^{(0)} \rangle = 0 \Rightarrow 0 = - \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \overbrace{\langle w_n^{(0)} | w_n^{(0)} \rangle}^1$$

$$\Rightarrow E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle$$

$$\bullet \boxed{\ell \neq n} \Rightarrow \langle w_\ell^{(0)} | w_n^{(0)} \rangle \text{ solo i sott. con } \ell = n$$

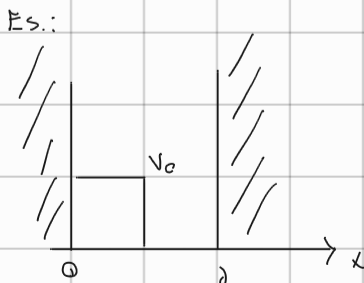
$$\Rightarrow (E_\ell^{(0)} - E_n^{(0)}) c_\ell = - \langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \cancel{\langle w_\ell^{(0)} | w_n^{(0)} \rangle}$$

$$\Rightarrow c_\ell = \frac{\langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle}{E_n^{(0)} - E_\ell^{(0)}}$$

$$w_n^{(1)} = \sum_{m \neq n} \frac{\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} w_m^{(0)}$$

(h) $\rightarrow$ AUTOSTATO CHE STO CORREGGENDO / AUTOS. DI CUI VOGLIO TROV. LA CORR.

(m) $\rightarrow$ AUTOST. CON CUI CORREGGO n



$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{2a}\right)$$

$$E_n^{(0)} = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2}$$

$$w_1^{(1)} = \sum_{n=1} \frac{\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle}{E_1^{(0)} - E_2^{(0)}} w_2^{(0)} + \frac{\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle}{E_1^{(0)} - E_3^{(0)}} w_3^{(0)} +$$

(bracket under the first term)

c_l

$$+ \frac{\langle \psi_n^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_n^{(0)}} \psi_n^{(0)} + \dots$$

$$c_2 = \frac{\langle \psi_2^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_2^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{2\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 4}{2m_e a^2}} = \frac{(4V_0/3\pi)}{-\frac{3\hbar^2 \pi^2}{2m_e a^2}}$$

$$c_3 = \frac{\langle \psi_3^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_3^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{3\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 9}{2m_e a^2}} = 0$$

$$c_4 = \frac{\langle \psi_4^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_4^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{4\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 16}{2m_e a^2}} = \frac{(-8V_0/15\pi)}{-\frac{15\hbar^2 \pi^2}{2m_e a^2}}$$

$$\psi_1 = \underbrace{\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)}_{\psi_1^{(0)}} - \underbrace{\left(\frac{8V_0 m_e a^2}{9\hbar^2 \pi^2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)}_{c_2 \psi_2^{(0)}} + \underbrace{\left(\frac{16V_0 m_e a^2}{225\hbar^2 \pi^2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{4\pi x}{a}\right)}_{c_4 \psi_4^{(0)}} + \dots$$

CORREZIONE al 2° ordine Ener.:

$$\boxed{\lambda^2} \quad \hat{H}^{(0)} \psi_n^{(2)} + \hat{H}^{(1)} \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)} \quad \text{Moltip. per } \psi_n^{(0)*}$$

$$\underbrace{\langle \psi_n^{(0)} | \hat{H}^{(0)} | \psi_n^{(2)} \rangle + \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle}_{\downarrow} = E_n^{(0)} \underbrace{\langle \psi_n^{(0)} | \psi_n^{(2)} \rangle}_{\downarrow} + E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle + E_n^{(2)} \langle \psi_n^{(0)} | \psi_n^{(0)} \rangle$$

$$\langle \psi_n^{(0)} | \psi_n^{(0)} \rangle = 1 \quad \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle = 0 \quad \langle \psi_n^{(0)} | \psi_n^{(2)} \rangle = 0$$

$$\Rightarrow E_n^{(2)} = \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle - \underbrace{E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle}_0$$

$$\Rightarrow \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle = \langle \psi_n^{(0)} | \sum_{m \neq n} c_m \psi_m^{(0)} \rangle = 0$$

$$\Rightarrow \boxed{E_n^{(2)} = \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle}$$

$$E_n^{(2)} = \langle w_n^{(0)} | \hat{H}' \sum_{m \neq n} \frac{c_m w_m^{(0)}}{E_n^{(0)} - E_m^{(0)}} \rangle = \sum_{m \neq n} c_m \langle w_n^{(0)} | \hat{H}' | w_m^{(0)} \rangle =$$

$$= \sum_{m \neq n} \frac{\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle \langle w_n^{(0)} | \hat{H}' | w_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} =$$

$$\Rightarrow E_n^{(2)} = \sum_{m \neq n} \frac{|\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

$n \Rightarrow$ STATO CHE CORREGGO

$m \Rightarrow$ UNO DEI STATI CON CUI CORREGGO

n

1) Suppongo n stato fond.: $E_n^{(0)} - E_m^{(0)} < 0$

$$\Rightarrow E_{SF}^{(2)} < 0$$

2) $\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle \rightarrow$ Tutti dello stesso ordine di grandezza

$\Rightarrow$ ciò che conta veramente è il denominatore, più lo stato con cui

voglio correggere è lontano da n più la correzione è piccola

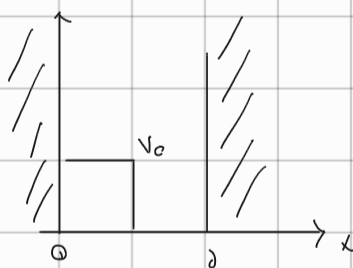
3) Se lo stato m sta sopra $n \Rightarrow E_n^{(0)} - E_m^{(0)} < 0 \Rightarrow$ lo stato m

spinge verso il basso n . Se $m < n \Rightarrow m$ spinge verso l'alto

$\Rightarrow$ TENDENZA dei LIV. ENERGETICI A RESPINGERSI

$\hookrightarrow$ No level-crossing theorem

Es.



$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$$E_n^{(0)} = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2}$$

$$E_1^{(2)} = \frac{|\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_2^{(0)}} + \frac{|\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_3^{(0)}} + \frac{|\langle w_4^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_4^{(0)}} + \dots$$

$$|\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = \frac{4 V_0}{3\pi}, \quad |\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = 0, \quad |\langle w_4^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = -\frac{8 V_0}{15\pi}$$

$$\Rightarrow E_1^{(2)} = \frac{\left(\frac{4 V_0}{3\pi}\right)^2}{\frac{\hbar^2 \pi^2}{2 m_e a^2} - \frac{4 \hbar^2 \pi^2}{2 m_e a^2}} + 0 + \frac{\left(-\frac{8}{15\pi} V_0\right)^2}{\frac{\hbar^2 \pi^2}{2 m_e a^2} - \frac{16 \hbar^2 \pi^2}{2 m_e a^2}} + \dots$$

$$\Rightarrow E_1 = \frac{\hbar^2 \pi^2}{2m_0 d^2} + \frac{V_0}{2} - \frac{32 V_0^2 m_0 d^2}{27 \hbar^2 \pi^4} - \frac{128 V_0^3 m_0 d^2}{3375 \hbar^2 \pi^4}$$

$E_1^{(2)}$

SISTEMI DEGENERI:

$$\left. \begin{aligned} \hat{H}^{(0)} w_{na}^{(0)} &= E_n^{(0)} w_{na}^{(0)} \\ \hat{H}^{(0)} w_{nb}^{(0)} &= E_n^{(0)} w_{nb}^{(0)} \end{aligned} \right\} \Rightarrow \langle w_{na}^{(0)} | w_{nb}^{(0)} \rangle = 0$$

SISTEMI DEGENERI di Dimensione (2)

$$w_n^{(0)} = \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \quad \alpha, \beta \in \mathbb{C}$$

$$\bullet E_n^{(1)} = \langle w_n^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle \Rightarrow \text{QUALE } w_n^{(0)} \text{ PRENDIAMO?}$$

$$\bullet E_n^{(0)} - E_m^{(0)} = E_{na}^{(0)} - E_{nb}^{(0)} = 0 \Rightarrow w_n^{(1)} \rightarrow +\infty \Rightarrow E_n^{(1)} \rightarrow +\infty$$

$$\boxed{\hat{H} w_n = E_n w_n} \quad \hat{H} = \hat{H}^{(0)} + \lambda \hat{H}^{(1)} \Rightarrow \begin{cases} w_n = w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots \\ E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \end{cases}$$

1° pot. in λ

$$[\hat{H}^{(0)} + \lambda \hat{H}^{(1)}] [w_n^{(0)} + \lambda w_n^{(1)}] = [E_n^{(0)} + \lambda E_n^{(1)}] [w_n^{(0)} + \lambda w_n^{(1)}]$$

$$\hat{H}^{(0)} w_n^{(0)} + \lambda \hat{H}^{(0)} w_n^{(1)} + \lambda \hat{H}^{(1)} w_n^{(0)} + \lambda^2 \hat{H}^{(1)} w_n^{(1)} = E_n^{(0)} w_n^{(0)} + \lambda E_n^{(0)} w_n^{(1)} + \lambda E_n^{(1)} w_n^{(0)} + \lambda^2 E_n^{(1)} w_n^{(1)}$$

$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}^{(1)} w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)} \quad \text{MOLTIPL. PER } w_{na}^{(0)*} \text{ e integ.}$$

$$\Rightarrow \langle w_{na}^{(0)} | \hat{H}^{(0)} | w_n^{(1)} \rangle + \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle$$

$$\hookrightarrow \langle \hat{H}^{(0)} w_{na}^{(0)} | w_n^{(1)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle$$

$$\Rightarrow E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle$$

$$\Rightarrow \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle, \quad w_n^{(0)} = \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)}$$

$$\langle w_{na}^{(0)} | \hat{H}^{(1)} | \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \rangle = E_n^{(1)} \langle w_{na}^{(0)} | \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \rangle$$

$$\Rightarrow \alpha \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_{na}^{(0)} \rangle + \beta \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_{nb}^{(0)} \rangle = \alpha E_n^{(1)}$$

POSSO FARE LA STESSA COSA PER $w_{nb}^{(0)}$ e ottengo:

$$\Rightarrow \alpha \langle w_{nb}^{(0)} | \hat{H}^{(1)} | w_{na}^{(0)} \rangle + \beta \langle w_{nb}^{(0)} | \hat{H}^{(1)} | w_{nb}^{(0)} \rangle = \beta E_n^{(1)}$$

$$\Rightarrow \begin{pmatrix} \langle w_{n_a}^{(0)} | \hat{H}' | w_{n_a}^{(0)} \rangle & \langle w_{n_a}^{(0)} | \hat{H}' | w_{n_b}^{(0)} \rangle \\ \langle w_{n_b}^{(0)} | \hat{H}' | w_{n_a}^{(0)} \rangle & \langle w_{n_b}^{(0)} | \hat{H}' | w_{n_b}^{(0)} \rangle \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{pmatrix} \hat{H}'_{aa} & \hat{H}'_{ab} \\ \hat{H}'_{ba} & \hat{H}'_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

MATRICE dell'op $\hat{H}'$ scritta sulla base w_{n_a} e w_{n_b}

↳ valore della comb. lin. $\alpha w_{n_a}^{(0)} + \beta w_{n_b}^{(0)}$

$\Rightarrow$ Se diagonali, la mat. trova le cor. al 1° ordine dell' E_n .

Autov. di $\hat{H}'$ scritta nel sottosp. degen. $\Rightarrow E_n^{(1)}$

Es.: δ_1, δ_2 (i due subsp. che trova)

$$\begin{array}{l} E_n^{(0)} \\ \swarrow \quad \searrow \\ E_1 = E_n^{(0)} + \delta_1 \\ E_2 = E_n^{(0)} + \delta_2 \end{array}$$

SE considero $\hat{H}'$ rimuovo la degenerazione

$$\Rightarrow \det(\delta I - \hat{H}') = 0 \quad H'_{ab} = H_{ba}^*$$

$$\det \begin{pmatrix} \delta_1 - H'_{aa} & -H'_{ab} \\ -H'_{ba} & \delta_1 - H'_{bb} \end{pmatrix} \Rightarrow \delta_1, \delta_2$$

$$\delta_1 \Rightarrow \begin{pmatrix} \delta_1 - H'_{aa} & -H'_{ab} \\ -H'_{ba} & \delta_1 - H'_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0 \Rightarrow \alpha_1, \beta_1 \Rightarrow \boxed{w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)}}$$

$\rightarrow$ Faccio uguale per $w_{n_2}^{(0)}$

$$E_n = E_n^{(0)} + \delta_1 \Rightarrow w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)}$$

$$\begin{array}{l} E_n^{(0)} \\ \swarrow \quad \searrow \end{array}$$

$$E_n = E_n^{(0)} + \delta_2 \Rightarrow w_{n_2}^{(0)} = \alpha_2 w_{n_a}^{(0)} + \beta_2 w_{n_b}^{(0)}$$

$$w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)} \Rightarrow \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \hookrightarrow \delta_1$$

$$w_{n_2}^{(0)} = \alpha_2 w_{n_a}^{(0)} + \beta_2 w_{n_b}^{(0)} \Rightarrow \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} \hookrightarrow \delta_2$$

NON È UNA CORREZ. di $w_n^{(0)}$

$w_{n_1}^{(0)}$ e $w_{n_2}^{(0)}$ SONO AUTOSTATI di $\hat{H}^{(0)}$

Quindi in questo modo per gli stati degeneri ottengo una correzione 1° ord. sull'En. che mi fa capire se i livelli si dividono o no e capisco il modo in cui gli stati imperturbati si mescolano tra loro.

CONOSCO A PRIORI (α_1, β_1) (α_2, β_2)

$$\begin{pmatrix} \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle & 0 \\ 0 & \langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle \end{pmatrix} \Rightarrow \text{la matrice è diagonale}$$

$\Rightarrow \psi_{n_1}^{(0)} \text{ e } \psi_{n_2}^{(0)} \Rightarrow$ BUONI STATI (quelli che diagonalizzano la matrice)

$$\left. \begin{aligned} \delta_1 &= \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle \\ \delta_2 &= \langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle \end{aligned} \right\} \text{TEORIA NON DEGENERE}$$

sono in grado di conoscere a priori i BUONI STATI!

TEOREMA:

cons. $\hat{O} = \hat{O}^\dagger \Rightarrow [\hat{O}, \hat{H}'] = 0$ se $\psi_{n_1}^{(0)}$ e $\psi_{n_2}^{(0)}$ sono Autostrati di $\hat{O}$ con

Autovalori distinti:

$$\left. \begin{aligned} \hat{O} \psi_{n_1}^{(0)} &= \xi \psi_{n_1}^{(0)} \\ \hat{O} \psi_{n_2}^{(0)} &= \nu \psi_{n_2}^{(0)} \end{aligned} \right\} \Rightarrow \boxed{H'_{12} = 0} \text{ la matrice è diag.}$$

Dim.: $[\hat{O}, \hat{H}] = 0$

$$\begin{aligned} \langle \psi_{n_1}^{(0)} | [\hat{O}, \hat{H}] | \psi_{n_2}^{(0)} \rangle &= 0 = \langle \psi_{n_1}^{(0)} | \hat{O} \hat{H}' | \psi_{n_2}^{(0)} \rangle - \langle \psi_{n_1}^{(0)} | \hat{H}' \hat{O} | \psi_{n_2}^{(0)} \rangle = \\ &= \langle \hat{O} \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle - \langle \psi_{n_1}^{(0)} | \hat{H}' | \hat{O} \psi_{n_2}^{(0)} \rangle = \xi \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle - \nu \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle = \\ &= (\xi - \nu) \underbrace{\langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle}_1 = 0 \\ &\quad \quad \quad \hookrightarrow = 0 \end{aligned}$$

■

δ^1

$$\begin{pmatrix} \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle & 0 \\ 0 & \underbrace{\langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle}_{\delta^2} \end{pmatrix}$$

DEGENERAZ. di ORDINE SUPERIORE:

$$\hat{H}^{(0)} w_{n0}^{(0)} = E_n^{(0)} w_{n0}^{(0)}$$

$$\hat{H}^{(0)} w_{nb}^{(0)} = E_n^{(0)} w_{nb}^{(0)}$$

⋮

$$\hat{H}^{(0)} w_{nk}^{(0)} = E_n^{(0)} w_{nk}^{(0)}$$

① IDENTICO SOPROSP. DEGENERE

$$w_n^{(1)} = a w_{n0}^{(0)} + \dots + k w_{nk}^{(0)}$$

$$\textcircled{2} \begin{pmatrix} H'_{00} & \dots & H'_{0k} \\ \vdots & & \vdots \\ H'_{k0} & \dots & H'_{kk} \end{pmatrix} \begin{pmatrix} a \\ b \\ \vdots \\ k \end{pmatrix} = E_n^{(1)} \begin{pmatrix} a \\ b \\ \vdots \\ k \end{pmatrix}$$

③ DIAGONALIZZO

STRUTTURA FINE dell' ATOMO di H: CORREZIONE RELATIVISTICA

$$\rightarrow w_n^{(0)} = w_{nl}^{(0)}$$

$$\hat{H}^{(0)} w_n^{(0)} = E_n^{(0)} w_n^{(0)} \rightarrow E_n^{(0)} = -\frac{R_H}{n^2}$$

$$\rightarrow \frac{\hat{p}^2}{2m} + V(r) = -\frac{\hbar^2}{2m} \Delta + V(r) \rightarrow \text{HAMILTONIANA di BOHR}$$

$$|E_n^{(0)}| \approx 10 \text{ eV}, \quad m_e c^2 \approx 500 \text{ KeV} \quad |E_n^{(0)}| \ll m_e c^2 \rightarrow \text{MOTO NON RELATIVISTICO}$$

RELATIVIST.

$$\vec{p} = \frac{m \vec{v}}{\sqrt{1 - (\frac{v}{c})^2}}$$

E cinetico rel.

$$K = \frac{m c^2}{\sqrt{1 - (\frac{v}{c})^2}} - m c^2 \Rightarrow \boxed{K \Leftrightarrow p} ?$$

$$\Rightarrow E = p^2 c^2 + m^2 c^4 = \frac{m^2 v^2}{1 - (\frac{v}{c})^2} c^2 + m^2 c^4 = \frac{m^2 v^2 c^2 + m^2 c^4 [1 - (\frac{v}{c})^2]}{1 - (\frac{v}{c})^2} =$$

$$= \frac{m^2 v^2 c^2 + m^2 c^4 - m^2 v^2 c^2}{1 - (\frac{v}{c})^2} = \frac{m^2 c^4}{1 - (\frac{v}{c})^2} = (K + m c^2)^2$$

$$\Rightarrow K = \sqrt{p^2 c^2 + m^2 c^4} - m c^2 = m c^2 \left[\sqrt{1 + \left(\frac{p}{m c}\right)^2} - 1 \right]$$

$$\Rightarrow \text{Limite non rel.} \Rightarrow p \ll m c \quad K = m c^2 \left[\sqrt{1 + \frac{p^2}{m^2 c^2}} - 1 \right] \approx$$

$$\approx m c^2 \left[1 + \frac{1}{2} \left(\frac{p}{m c}\right)^2 - \frac{1}{8} \left(\frac{p}{m c}\right)^4 + \dots - 1 \right] \Rightarrow \boxed{K = \frac{p^2}{2m} - \frac{p^4}{8 m^3 c^2} + \dots}$$

$$\left[\begin{aligned} \hat{H}'_r &= - \frac{\hat{p}^4}{8m^3 c^2} \end{aligned} \right]$$

(b) $\Rightarrow l = 0, \dots, n-1 \Rightarrow$ TANTI STATI degeneri

$$\hat{H}'_r \Rightarrow [\hat{H}'_r, \hat{L}^2] = [\hat{H}'_r, \hat{L}_z] = 0 \quad (\text{posso applicare Teorema per stati deg.})$$

$$\hat{L}^2 \Rightarrow w_{nlm} (\text{AUTOST.}) \leftarrow \hat{L}_z$$

$$\hat{L}^2 w_{nlm} = \hbar^2 l(l+1) w_{nlm} \quad (\text{non degeneri per } l \text{ per } \hat{L}^2)$$

$$[\hat{O}, \hat{H}'] = [\hat{L}^2, \hat{H}'_r] = 0 \quad \text{ho trovato l'op. che mi serve con stati non degeneri.}$$

$\langle w_{nlm} | \hat{H}'_r | w_{nlm} \rangle \rightarrow$ diagonale le autof. dell'atomo di H sono già: BUONI STATI
 $\Rightarrow$ POSSO USARE TEOR. NON DEG.

$$E_n^{(1)} = - \frac{1}{8m^3 c^2} \langle w_n^{(0)} | \hat{p}^4 | w_n^{(0)} \rangle = - \frac{1}{8m^3 c^2} \langle w_n^{(0)} | \hat{p}^2 \hat{p}^2 | w_n^{(0)} \rangle = \frac{1}{8m^3 c^2} \langle \hat{p}^2 w_n^{(0)} | \hat{p}^2 w_n^{(0)} \rangle$$

$$\Rightarrow \frac{\hat{p}^2}{2m} w_n^{(0)} + V w_n^{(0)} = E_n^{(0)} w_n^{(0)}$$

$$\left[\begin{aligned} \hat{p}^2 w_n^{(0)} &= 2m(E_n^{(0)} - V) w_n^{(0)} \end{aligned} \right]$$

$$= - \frac{1}{8m^3 c^2} \langle 2m(E_n^{(0)} - V) w_n^{(0)} | 2m(E_n^{(0)} - V) w_n^{(0)} \rangle = - \frac{1}{8m^3 c^2} \langle m^2 (E_n^{(0)} - V)^2 \rangle$$

$$E_n^{(1)} = - \frac{1}{2\hbar c^2} \langle (E_n^{(0)} - V)^2 \rangle = - \frac{1}{2\hbar c^2} \langle E_n^{(0)2} + \hat{V}^2 - 2E_n^{(0)} \hat{V} \rangle$$

$$= - \frac{1}{2\hbar c^2} \left[\langle E_n^{(0)2} \rangle - 2E_n^{(0)} \langle \hat{V} \rangle + \langle \hat{V}^2 \rangle \right] =$$

$$= - \frac{1}{2\hbar c^2} \left[\left(- \frac{\partial \psi_0}{\partial r} \right)^2 + 2 \left(\frac{\partial \psi_0}{\partial r} \right) \left(\frac{e^2}{4\pi\epsilon_0} \right) \left(\frac{1}{r} \right) + \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left(\frac{1}{r^2} \right) \right]$$

$$\left[\left(\frac{1}{r} \right) \Rightarrow \right] \text{TEO del VIRIALE} \quad V(r) = r^K \Rightarrow 2 \langle E_K \rangle = K \langle V \rangle$$

$$V(r) = - \frac{e^2}{4\pi\epsilon_0 r} \quad K = -1 \Rightarrow 2 \langle E_K \rangle = - \langle V \rangle, \quad E = \langle K \rangle + \langle V \rangle = - \frac{\langle V \rangle}{2} + \langle V \rangle = \frac{\langle V \rangle}{2}$$

$$\Rightarrow \Delta V = 2E \Rightarrow -\frac{e^2}{4\pi\epsilon_0} \Delta \frac{1}{r} = 2E_n^{(0)} = 2\left(-\frac{e^2}{n^2}\right) = 2\left(-\frac{me^4}{(4\pi\epsilon_0)^2 \hbar^2} \cdot \frac{1}{n^2}\right)$$

$$\Delta \frac{1}{r} = \frac{\hbar c^2}{4\pi\epsilon_0 \hbar^2} \cdot \frac{1}{n^2} = \frac{1}{n^2 \partial_0^2}, \quad \Delta \frac{1}{r^2} = \int_{\mathbb{R}^3} w_{nlm}^{(0)*} \frac{1}{r^2} w_{nlm}^{(0)} \sin\theta d\theta d\phi dr =$$

$$= \frac{1}{(l+\frac{1}{2}) n^3 \partial_0^2}$$

$$E_n^{(1)} = -\frac{1}{2mc^2} \left[\frac{\partial w}{\partial r} + 2\left(\frac{\partial w}{\partial r}\right) \left(\frac{e^2}{4\pi\epsilon_0}\right) \left(\frac{1}{n^2 \partial_0^2}\right) + \left(\frac{c^2}{4\pi\epsilon_0}\right)^2 \frac{1}{(l+\frac{1}{2}) n^3 \partial_0^2} \right] =$$

$$E_n^{(1)} = -\frac{(E_n^{(0)})^2}{2mc^2} \left[\frac{4n}{l+\frac{1}{2}} - 3 \right]$$

→ la degener. per l è rimossa

$$\Rightarrow \frac{E_n^{(1)}}{E_n^{(0)}} \approx \frac{E_n^{(0)}}{mc^2} \approx 10^{-4}$$

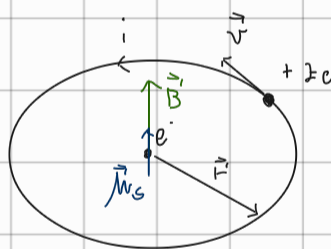
$\begin{matrix} \nearrow 10^{-18} \\ \searrow 10^{-51} \end{matrix}$

STRUTTURA FINE dell'ATOMO di H: SPIN ORBITA

$$\hat{H}_{so} = \left(\frac{e^2}{8\pi\epsilon_0}\right) \frac{1}{n^2 l^2 r^3} \hat{\vec{S}} \cdot \hat{\vec{L}}$$

$$\Delta E_{so} = -\vec{M}_S \cdot \vec{B}$$

Doppio di: SPIN ORBITA



$$\hat{H}^{(0)} = -\frac{\hbar^2}{2m} \Delta - V(r)$$

$E_n^{(0)}$

3v. deg. $\nearrow$

$n=3, l=0,1,2 \quad s=1/2 \Rightarrow j=5/2, 3/2, 1/2$

2v. deg. $\nearrow$

$n=2, l=0 \quad s=1/2 \Rightarrow j=3/2, 1/2$

$n=1, l=0, s=1/2 \Rightarrow j=1/2$

($\hat{H}^{(0)}$ crea un sottosp. degenerato con dimens. l)

$$|w_{nlm}\rangle = |n, l, m_l, s, m_s\rangle, \quad |n, j, m, l, s\rangle$$

↳ BASE DISACCOPIATA

↳ BASE ACCOPIATA

→ $w_{nlm} \Rightarrow$ STATI IMPERIGURATI nella BASE DISACCOPIATA

$$\Gamma_n=2 \rightarrow \left[\langle n, m, m | \hat{H}'_{so} | n, m, m \rangle \right] 2 \times 2, \quad \Gamma_n=3 \rightarrow \left[\langle n, m, m | \hat{H}'_{so} | n, m, m \rangle \right] 3 \times 3$$

→ Divergenza dei valori da risolvere

Posso trovare un metodo più semplice $\Rightarrow$ Trovo $\hat{O}$ t.c.:

$$[\hat{O}, \hat{H}'_{so}] = 0, [\hat{H}'_{so}, \hat{J}^2] = [\hat{H}_{so}, \hat{J}_z] = 0 \quad |n, j, m_j, l, s\rangle$$

↑

$$E_n^{(1)} = \langle n, j, m_j, l, s | \hat{H}'_{so} | n, j, m_j, l, s \rangle$$

$$\Rightarrow E_n^{(1)} = \left(\frac{e^2}{8\pi\epsilon_0} \right) \frac{1}{m^2 c^2} \langle n, j, m_j, l, s | \frac{\vec{S} \cdot \vec{L}}{r^3} | n, j, m_j, l, s \rangle$$

$$\hat{J}^2 = (\hat{L} + \hat{S})(\hat{L} + \hat{S}) = \hat{L}^2 + \hat{S}^2 + 2\hat{L} \cdot \hat{S} \Rightarrow \hat{L} \cdot \hat{S} = \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$$

$$E_n^{(1)} = \frac{e^2}{16\pi\epsilon_0 m^2 c^2} \langle n, j, m_j, l, s | \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{r^3} | n, j, m_j, l, s \rangle$$

$$\text{Proprietà } \Rightarrow \langle \hat{A} \cdot \hat{B} \rangle = \langle \hat{A} \rangle \langle \hat{B} \rangle \Leftrightarrow [\hat{A}, \hat{B}] = 0$$

$$\left[\frac{1}{r^3}, \hat{J}^2 \right] = \left[\frac{1}{r^3}, \hat{L}^2 \right] = \left[\frac{1}{r^3}, \hat{S}^2 \right] = 0$$

$$E_n^{(1)} = \frac{e^2}{16\pi\epsilon_0 m^2 c^2} \langle \hat{J}^2 - \hat{L}^2 - \hat{S}^2 \rangle \langle \frac{1}{r^3} \rangle$$

$$\hookrightarrow \hbar^2 j(j+1) - \hbar^2 l(l+1) - \hbar^2 s(s+1)$$

$$\left\langle \frac{1}{r^3} \right\rangle = \int_{\mathbb{R}} \dots = \frac{1}{l(l+\frac{1}{2})(l+1) n^3 a_0^3}$$

$$E_n^{(1)} = \frac{e^2}{16\pi\epsilon_0 m^2 c^2} \frac{\hbar^2 [j(j+1) - l(l+1) - \overbrace{s(s+1)}^{3/4}]}{l(l+\frac{1}{2})(l+1) n^3 a_0^3}$$

$$s, l=0 \Rightarrow j=s \Rightarrow \frac{0}{0}$$

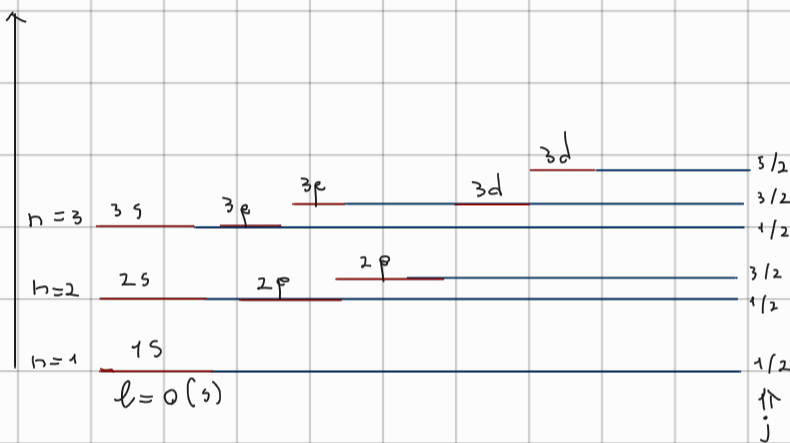
$$\Rightarrow E_{n,so}^{(1)} = \frac{(E_{n,so}^{(0)})^2}{m c^2} \left\{ \frac{n [j(j+1) - l(l+1) - 3/4]}{l(l+1)(l+1/2)} \right\}$$

$$\frac{E_{n,so}^{(1)}}{E_{n,so}^{(0)}} \approx \frac{E_{n,so}^{(0)}}{m c^2} \approx 10^{-4}$$

eq. d: strett. fine

$$E_{n, sf}^{(1)} = E_{n, EL}^{(1)} + E_{n, so}^{(1)} = \frac{(E_n^{(0)})^2}{2mc^2} \left[3 - \frac{4n}{j + \frac{1}{2}} \right]$$

$$E_n \approx E_n^{(0)} + E_{n, sf}^{(1)}$$

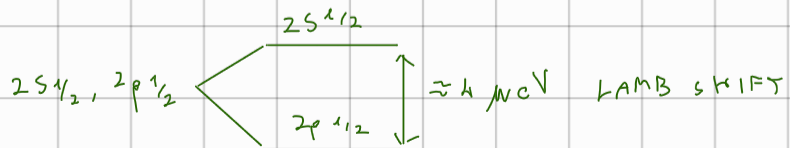


→ U-C 2s e 2p^{1/2} alobizano

lo stesso E_n è conf. da

DIRAC

1954 Lamb



Dovuto a fluttuaz. del campo E.M. nel vuoto